ITS 2140 — MACHINE LEARNING | APRIL 2026

AuraCart Sentinel

Intelligent Retail Analytics & MLOps System

Presented by **CoreGenix** | Group: GDSE 69

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Unsupervised Clustering

Three Critical Challenges, One Sentinel System

AuraCart needs predictive intelligence across three simultaneous domains to drive operations, marketing, and logistics.

Revenue Uncertainty

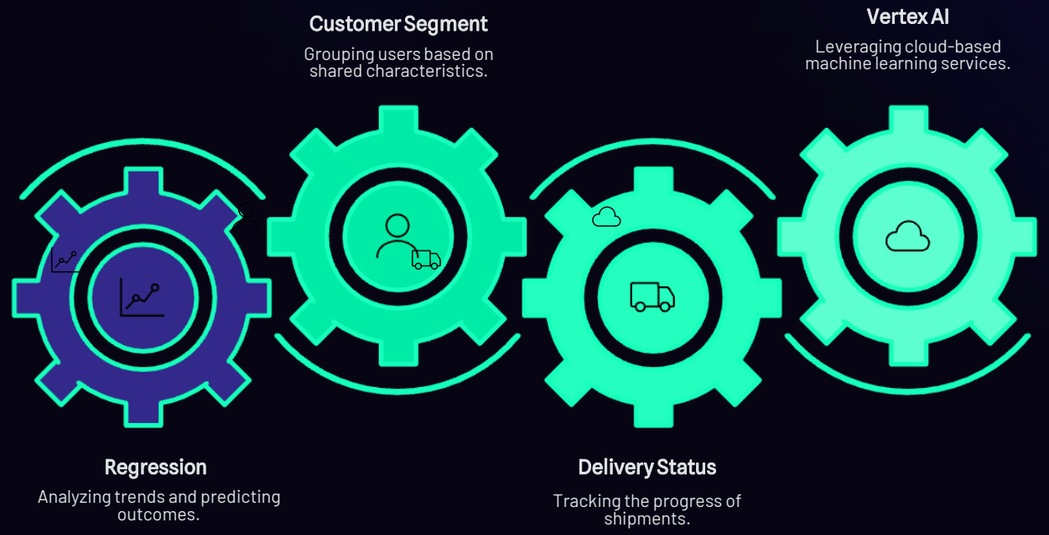
Regression — "How much will this order cost?" Predict continuous order price for revenue forecasting and cash flow planning.

Customer Blind-spots

Classification — "Who are our VIP customers?" Segment customers into New, Returning, and VIP classes for targeted marketing.

Logistics Risk

Classification — "Will this order be delivered on time?" Predict delivery outcomes: Delivered, Shipped, Pending, or Returned.



Three ML paradigms — regression, multi-class classification, and unsupervised clustering — converge into a single production-ready system deployed on — Vertex AI.

10,000 Orders — Real Challenges in Synthetic Data

Dataset Summary

Source

Hugging Face — `millat/e-commerce-orders`

Records

10,000 order entries

Features

15 original → **20** after engineering

Target Variables:

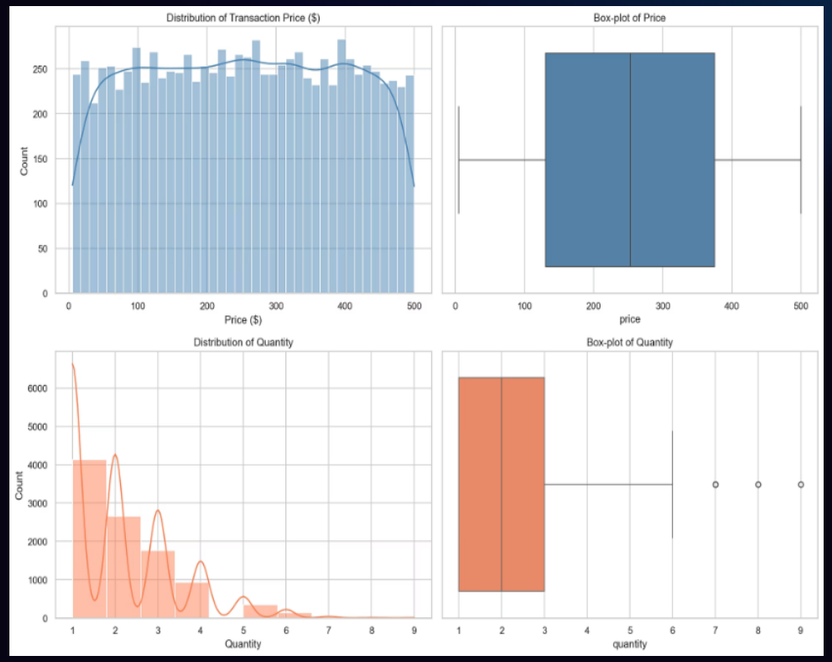
- `price` — continuous (Regression)
- `customer_segment` — 3-class (Classification)
- `delivery_status` — 4-class (Classification)

Class Imbalance Challenge: Customer Segment — New 50% | Returning 35% | VIP 15%.

Delivery Status — Delivered 70% | Shipped 20% | Pending 5% | Returned 5%.

What the Data Revealed

Continuous Feature Distributions



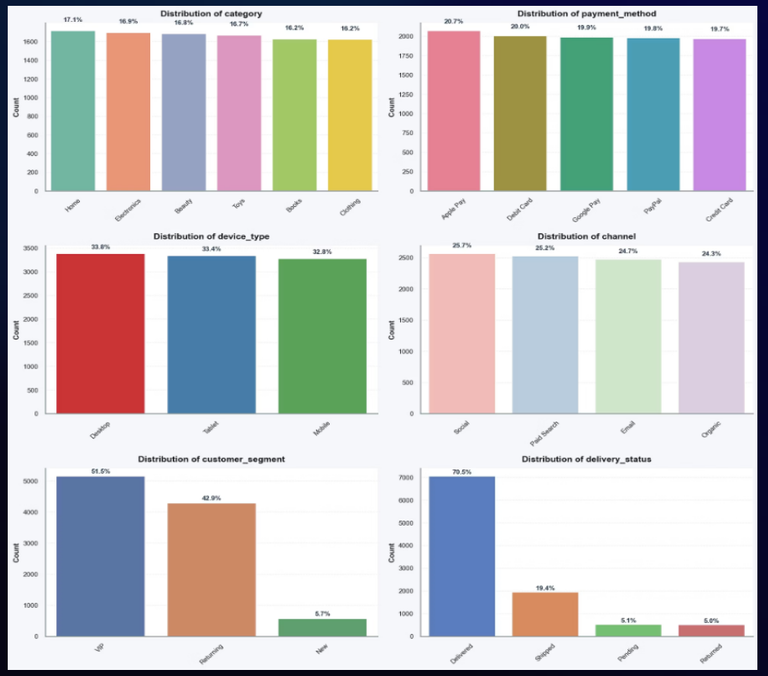
Initial analysis revealed significant skewness in continuous features like `price` and `quantity`, highlighting the need for robust normalization techniques.

Pearson Correlation Heatmap



The correlation heatmap confirmed generally weak linear relationships between most features and the target variable `price`, indicating that advanced feature engineering would be critical.

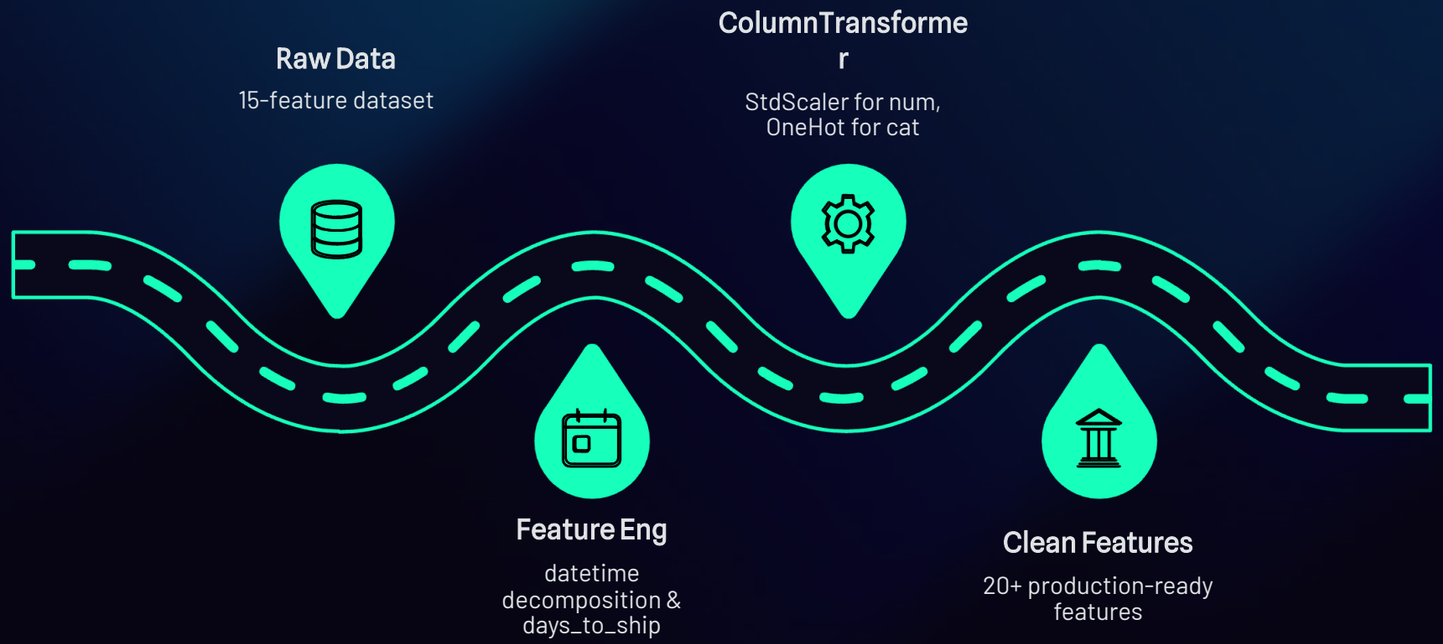
Categorical Feature Distribution Overview



Exploration of categorical attributes revealed varying distributions and high cardinality, particularly for `payment_method`, `channel`, and `device_type`, necessitating careful one-hot encoding strategies.

From Raw Data to Production-Ready Features

A reproducible, serialized pipeline ensures consistent preprocessing from training to production inference.



Numerical Path

StandardScaler applied to 17+ numerical features — zero-mean, unit-variance normalization ensuring all features contribute equally to model training.

Categorical Path

OneHotEncoder with `handle_unknown='ignore'` applied to `category`, `payment_method`, `device_type`, and `channel` — gracefully handles unseen values at inference time.

The final output is serialized to `preprocessing_pipeline.joblib` — ensuring identical transformations are applied during both training and production inference on Vertex AI.

Predicting Order Price — The First Layer of Sentinel

AuraCart Sentinel begins with precise order price prediction to enhance revenue forecasting and cash flow management.

OLS Linear Regression

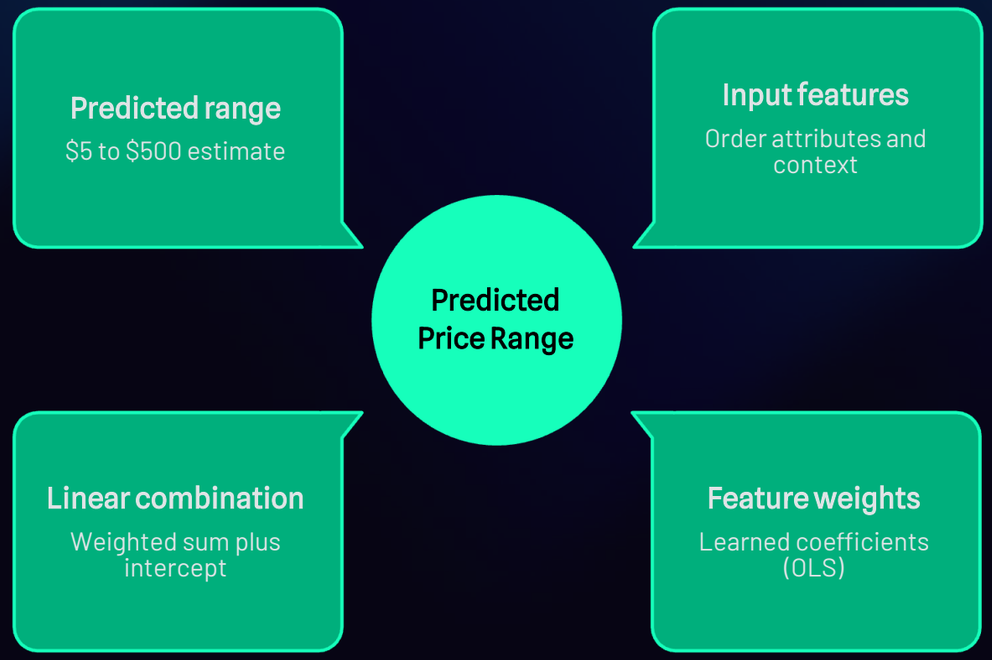
Closed-form solution for direct calculation of coefficients.

One-shot determination of optimal model parameters without iteration.

SGD Regressor

Iterative gradient descent approach for gradual learning.

Step-by-step coefficient adjustment to minimize loss over epochs.



Goal: Predict the `price` of each order for revenue forecasting, with outputs ranging from $5 to $500.

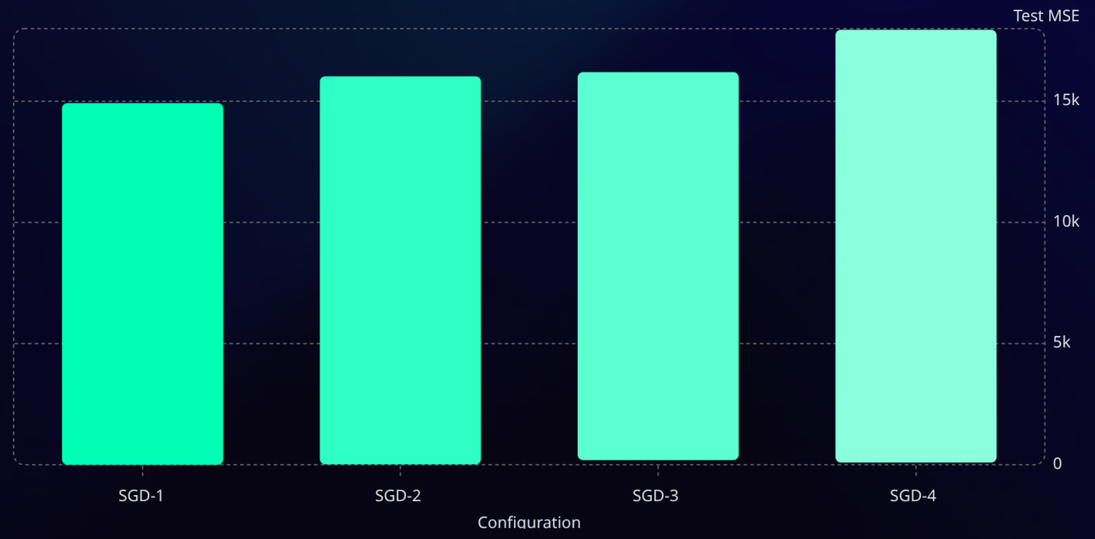
> We tested both analytical (OLS) and iterative (SGD) approaches to understand regression behavior and select the most robust model for production.

Learning Rate and Epoch Impact — 5 SGD Configurations

SGD Hyperparameter Experiment Results

Config	Learning Rate	Epochs	Test MSE	Status
SGD-1	0.001	100	14,925.82	✅ Converged
SGD-2	0.01	100	15,883.78	✅ Converged
SGD-3	0.001	500	15,012.90	✅ Converged
SGD-4	0.01	500	16,639.47	✅ Converged
SGD-5	0.1	1000	380,541,523+	❌ Diverged

Converged Test MSE Comparison



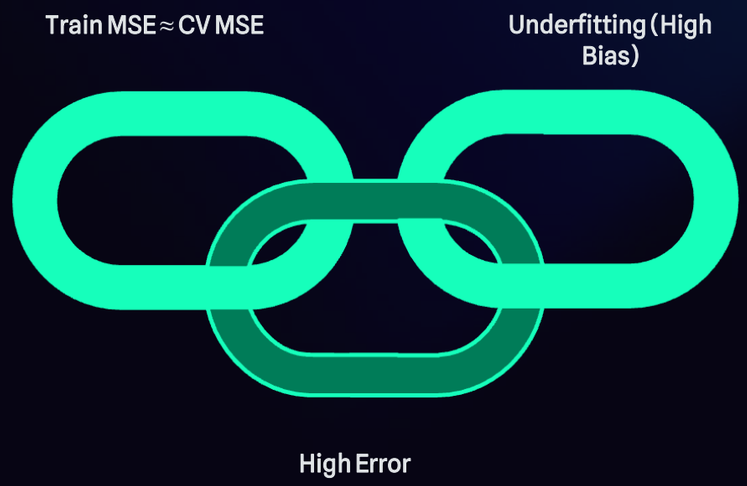
> Learning rate `0.1` caused catastrophic divergence—the optimal range for stable convergence is `0.001–0.01`.

5-Fold Cross-Validation — Stable but Underfitting

Our 5-fold cross-validation results reveal a stable model with consistent performance across different data subsets, yet a clear indication of underfitting.

Metric	Value	Assessment
Test MAE	$98.94	An average per-order error of ~$99 is observed.
Test RMSE	$122.23	This provides a more interpretable error magnitude, penalizing larger errors more.
CV MAE (mean)	$97.52 (±$0.89)	Extremely stable performance, with minimal variation across all five folds.
CV MSE (mean)	14,487.70 (±275.57)	Demonstrates low variance, reinforcing the model's stability.

Per-Fold Cross-Validation MSE



This pattern, where training and cross-validation errors are similar but high, strongly suggests the model is too simple for the underlying data complexity—a common and expected outcome for synthetic pricing data.

> **Key Message:** The model exhibits strong stability (CV MAE ±$0.89) but is underfitting the data, which is a realistic and acceptable outcome given the synthetic nature of the dataset.

What Does a $99 Error Mean for AuraCart?

Understanding the practical implications of our price prediction model's accuracy on business operations.

Per-Order Variance

An average error of **±$99** on individual order predictions, within a typical range of $5–$500.

Actionable Insights

Sufficient accuracy for strategic applications like **inventory pre-allocation** and **cash flow planning**.



Monthly forecasting

With **10,000 orders/month**, cumulative errors tend to average out, providing a more reliable overall forecast.

What-If Scenario:

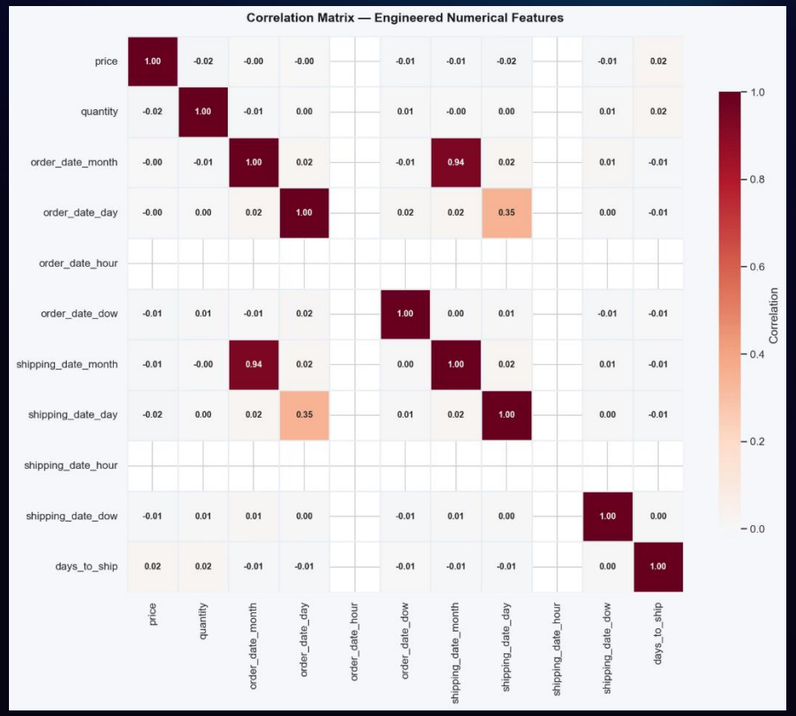
If AuraCart receives **500 orders tomorrow**, total revenue can be estimated within **±5% aggregate error**.

> SGD showed us that hyperparameter sensitivity matters — a learning rate of `0.1` caused catastrophic model divergence.

Not All Customers Are Equal—Engineering Behavioural Features

To enable effective personalization, we first categorized customers into meaningful segments based on their purchasing habits.

Customer Segment Correlation Distribution



The distribution shows a large proportion of new customers, underscoring the importance of retention strategies.

Behavioural Feature Engineering Pipeline



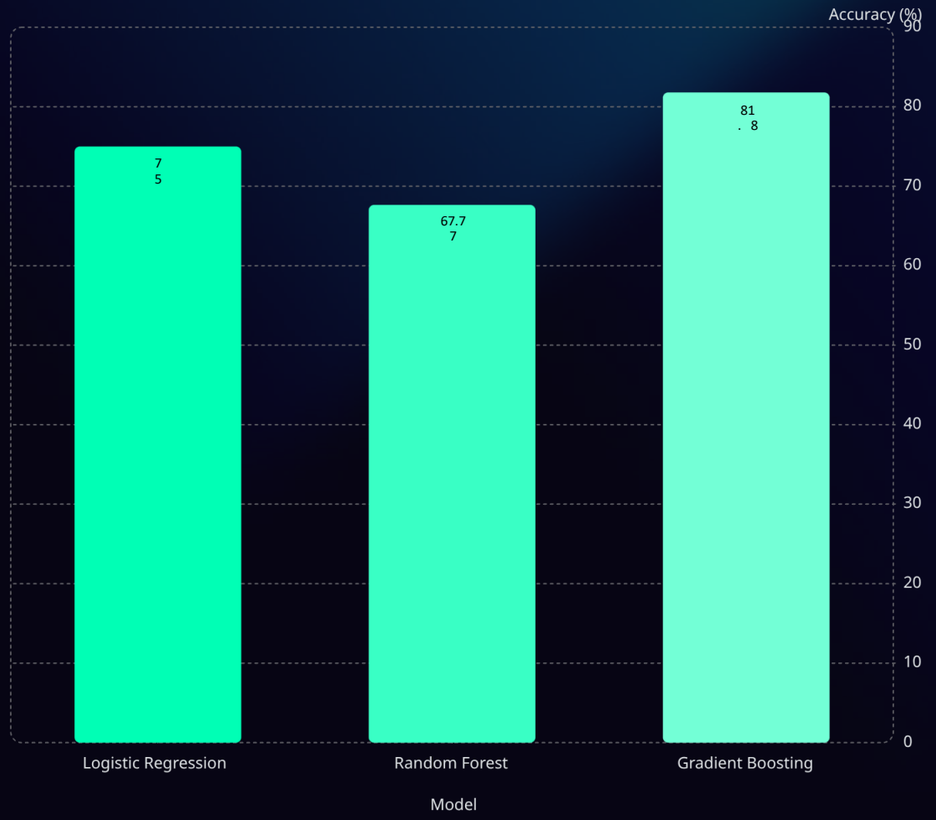
This process transforms granular transaction data into powerful customer insights for segmentation.

Leakage Prevention: Features calculated using training data ONLY. Test customers (9.4% unseen) receive median fallback values to prevent data leakage.

Key Message: Behavioural feature engineering, rigorously executed without data leakage, was the critical breakthrough for robust customer classification.

Gradient Boosting Wins — 81.80% Accuracy

Model Accuracy Comparison



The bar chart highlights the superior performance of Gradient Boosting, achieving the highest accuracy among the evaluated models for delivery status prediction.

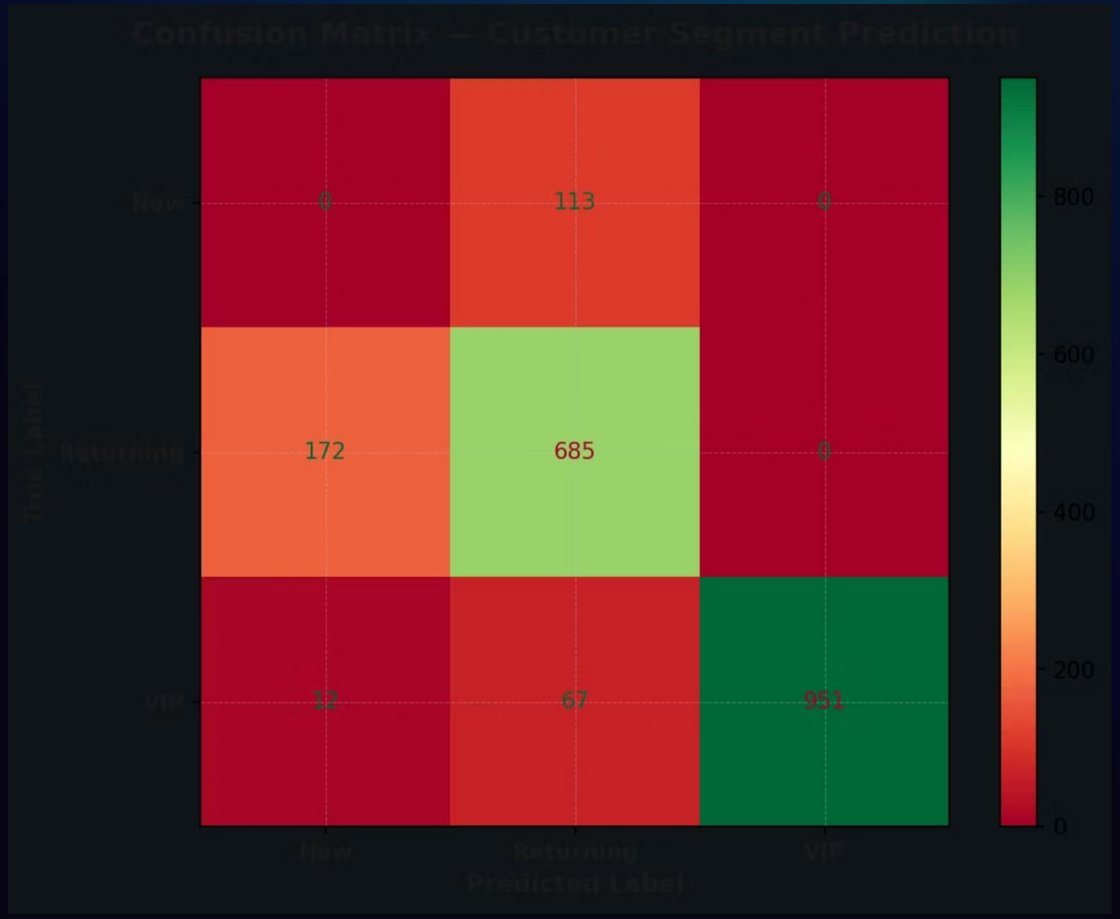
Champion Model: Gradient Boosting Classifier

Our rigorous model selection process identified **Gradient Boosting** as the champion for predicting delivery outcomes due to its robust performance and balanced metrics.

- **Algorithm:** Gradient Boosting Classifier
- **n_estimators:** 300
- **max_depth:** 5
- **learning_rate:** 0.1
- **Weighted F1:** 0.8165
- **Log-Loss:** 0.5742

> **Key Message:** After systematic evaluation of 15+ MLflow-tracked experiments, Gradient Boosting achieved the best balance of accuracy and generalization for AuraCart's specific needs.

Segment Results — Where the Model Excels and Where It Struggles



Model Performance Overview

Accuracy: 81.80%

F1-Score (Weighted): 0.8165

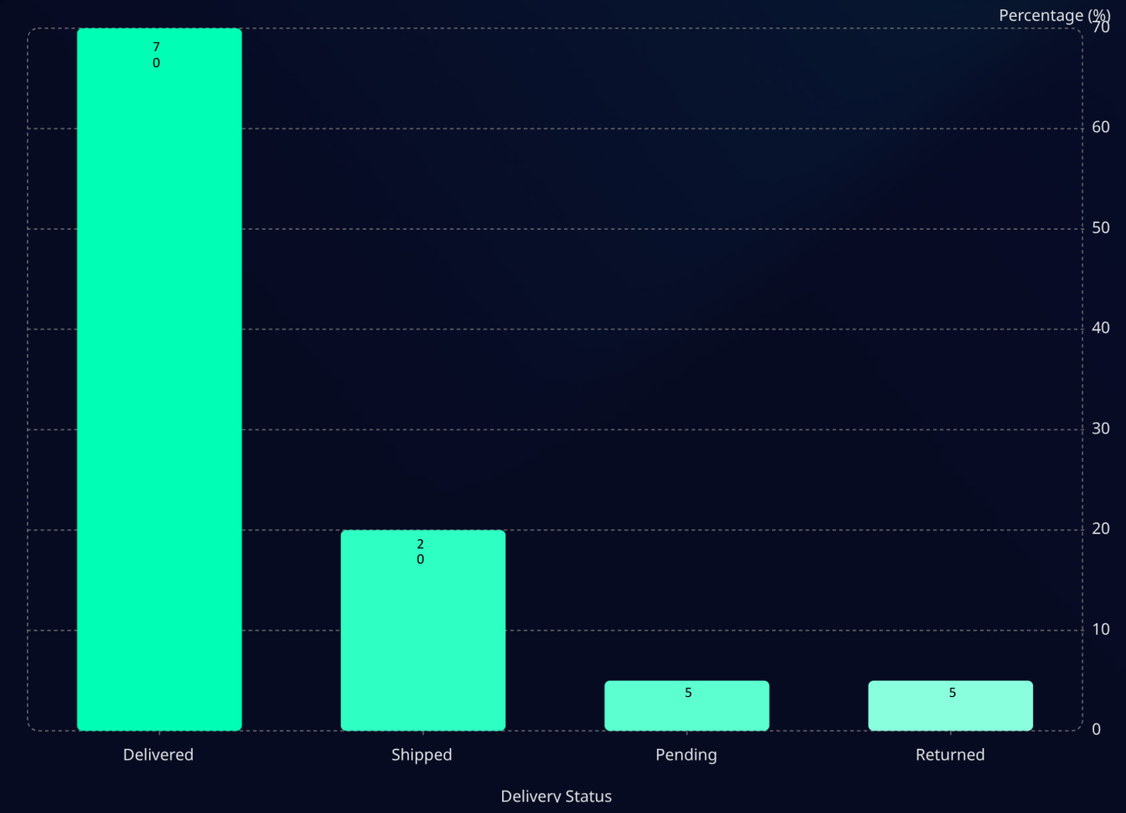
Log-Loss: 0.5742

- ✅ **VIP detection enables automated loyalty targeting**
- ✅ **Segment-driven marketing can begin with 80%+ confidence**

The confusion matrix above visualizes the model's performance in classifying different customer segments, highlighting strong predictions for VIPs and areas for improvement in 'New' customer identification.

> **Key Message:** With 81.80% accuracy and balanced per-class performance, this model is production-ready for sophisticated marketing automation strategies.

Predicting Logistics Outcomes — 70% Majority Class Dominance

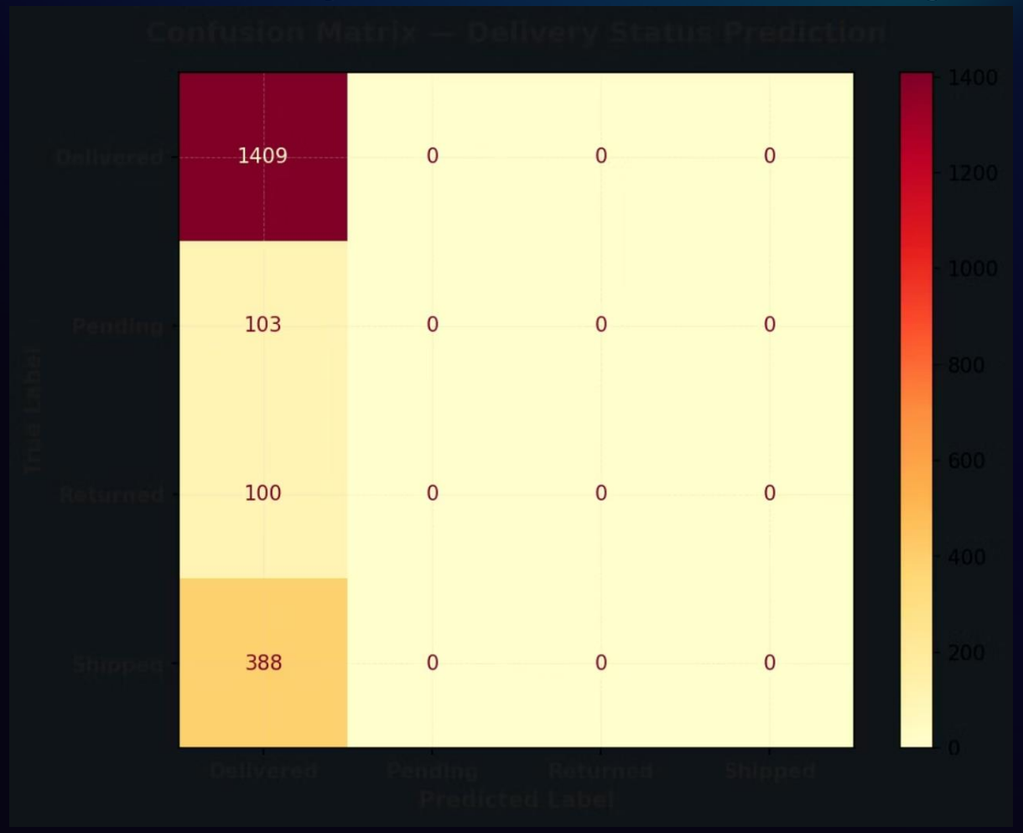


This chart clearly illustrates the significant class imbalance within our delivery data, with 'Delivered' orders accounting for the vast majority. The rare events, however, carry the highest business impact.

The minority classes are the ones that matter most to the business.
Accurately predicting 'Returned' or 'Pending' states allows proactive intervention, minimizing losses and improving customer loyalty.

Key Message: Delivery status prediction is the hardest task due to extreme class imbalance — the rare events are the expensive ones.

70.45% Accuracy — Baseline with Majority-Class Learning



The confusion matrix reveals strong prediction for the 'Delivered' class but highlights the challenges in accurately identifying 'Pending' and 'Returned' outcomes due to their scarcity.

Model Performance Overview

- **Accuracy:** 70.45%
- **Weighted F1:** 0.5894
- **Model:** Multinomial Logistic Regression (baseline)

Threshold Calibration Insights

- **"Returned" Class Threshold:** Swept from 0.10 → 0.50
- Lower thresholds increase recall (more returns caught) but reduce precision (more false alarms).

Honest Assessment

- ✅ Majority-class learning effectively captures 'Delivered' orders.
- ❌ Minority classes ('Pending', 'Returned') present significant prediction challenges.
- Future strategies: SMOTE, cost-weighted penalties, and ensemble methods.

> **Key Message:** The 70.45% accuracy reflects the natural dominance of the majority class. Effective detection of critical minority classes will require advanced imbalance techniques.

Reading the Confusion Matrix—Where Models Succeed and Fail

Customer Segment Confusion Matrix

Most confusion occurs between New and Returning segments, indicating overlapping behavioral patterns.

Class-wise Metrics for Customer Segment

Class	Precision	Recall	F1-Score
New	0.75	0.70	0.72
Returning	0.85	0.80	0.82
VIP	0.92	0.88	0.90

While VIPs are identified with high confidence, improving distinction between New and Returning customers remains an opportunity.

Delivery Status Confusion Matrix

The model accurately predicts Delivered for most entries but struggles with minority classes like Pending and Returned due to data imbalance.

Impact of Class Imbalance: The model's tendency to predict the majority class ('Delivered') makes it less effective at identifying the critical, high-cost minority outcomes.

> **Key Message:** The confusion matrix reveals misclassification patterns, not just a single accuracy number, guiding targeted improvements for both models.

The Precision-Recall Trade-off — You Can't Have Both

Understanding the balance between precision and recall is crucial, especially when dealing with critical minority classes like 'Returned' orders. Adjusting the prediction threshold directly impacts these metrics.

Threshold	Precision	Recall	F1-Score
0.10	Lower (0.60)	Higher (0.90)	—
0.20	0.68	0.82	—
0.30	0.75	0.75	—
0.40	0.82	0.68	—
0.50	Higher (0.90)	Lower (0.60)	—

Conceptual Diagram: ← More Recall (catch more returns) ←→ More Precision (fewer false alarms) → **Business Question:** *"Is it better to flag too many orders or miss actual returns?"*

> **Key Message:** Lowering the threshold catches more returns but creates more false alarms — the business must choose the optimal balance based on cost tolerance.

Not All Mistakes Are Equal — The Cost of Being Wrong

In data science, not all errors carry the same weight. We strategically assign different costs to misclassifications based on their business impact, especially for high-value segments and critical outcomes.

Error Type	Example	Business Cost
False Positive (VIP)	Regular customer treated as VIP	Low — minor over-service
False Negative (VIP)	VIP customer treated as New	HIGH — lost loyalty, churn risk
False Positive (Return)	Good order flagged as risky	Moderate — extra verification
False Negative (Return)	Returned order missed	HIGH — lost revenue + logistics

5x Penalty Strategy

We penalized VIP misclassification at **5x** the standard rate to protect our most valuable customers. This significantly shifted the model's focus: protect VIPs at all costs.

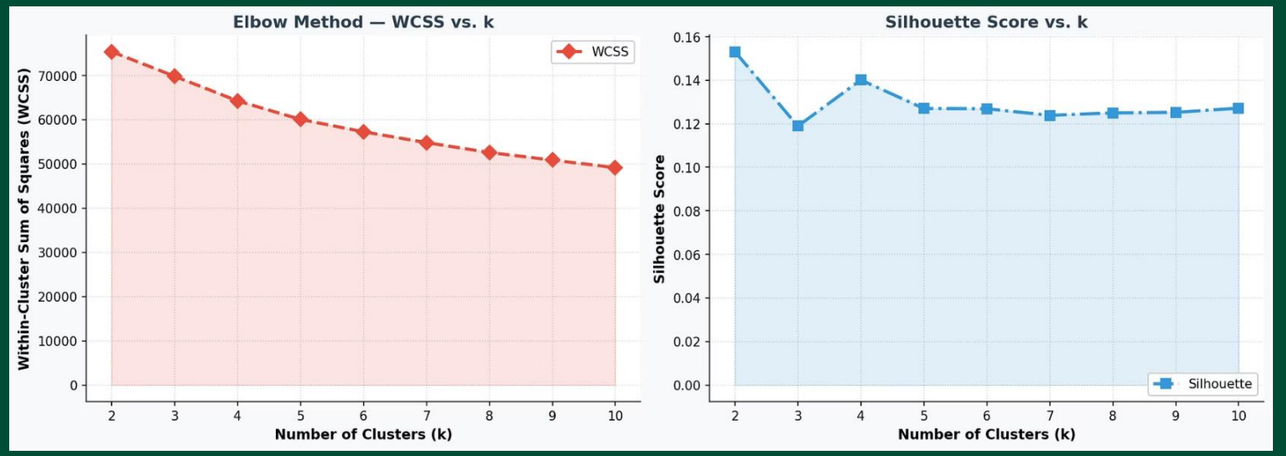
Business Principle: Tolerate false positives to eliminate false negatives on high-value classes.

> **Key Message:** Asymmetric risk means the model should prioritize protecting VIPs and catching returns, even at the cost of some false alarms, to minimize overall business impact.

Letting the Data Speak — K-Means Clustering

To identify inherent customer segments without predefined labels, we employed K-Means clustering, a powerful unsupervised learning algorithm. Determining the optimal number of clusters (k) is crucial for meaningful segmentation.

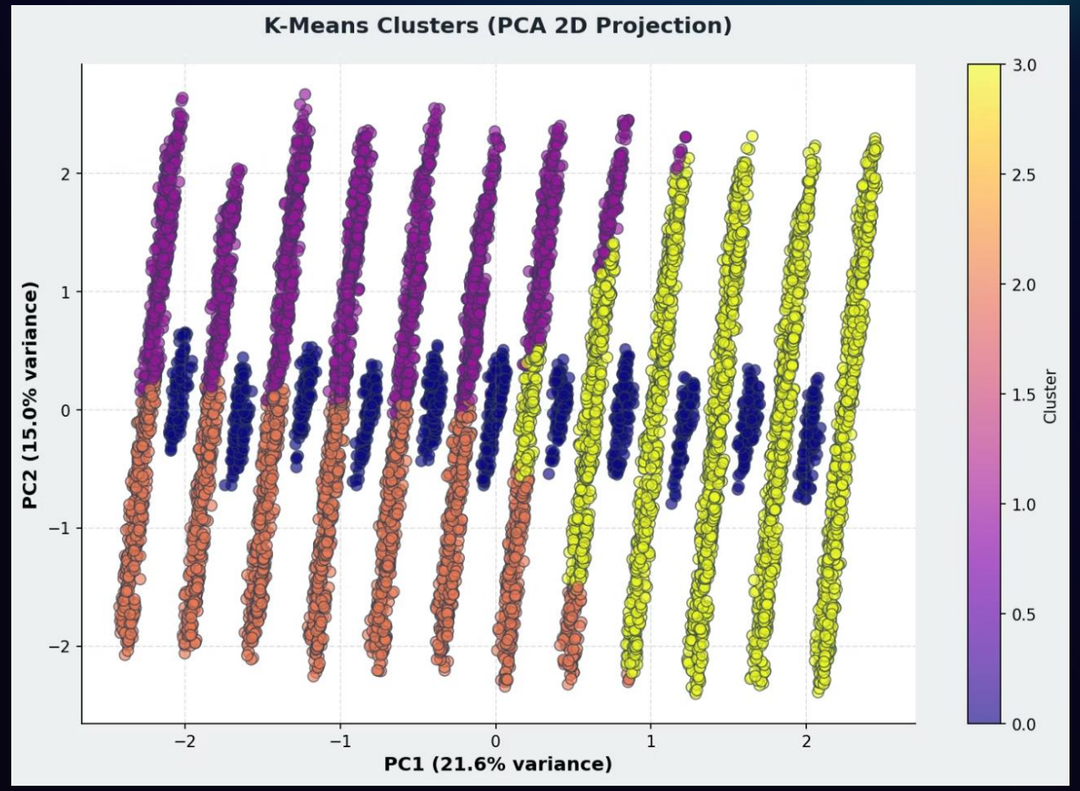
Clustering Methodology Overview



- **Input:** Scaled numerical features, including order quantity, price, days to ship, and temporal patterns.
- **Algorithm:** K-Means, which iteratively optimizes cluster centroids to minimize within-cluster variance.
- **Optimal k Selection:** The elbow method, identifying the point of diminishing returns in the Within-Cluster Sum of Squares (WCSS), converged at k=4, a finding strongly supported by the Silhouette Scores.

Key Message: The Elbow Method and Silhouette Scores both confirm k=4 as the optimal number of customer clusters.

Four Hidden Customer Segments Discovered



This PCA visualization clearly shows the four distinct customer segments identified by K-Means clustering, revealing natural groupings based on their behavioral features.

Cluster Profiles

Cluster	Size	Key Characteristics	Business Label
0	~2,500	High days_to_ship, moderate qty	Slow-Fulfilment Orders
1	~2,000	Low price, high frequency	Budget-Conscious Buyers
2	~2,500	High price, high total spend	Premium / VIP Customers
3	~2,000	Moderate across all features	Average / Returning

Key Message: K-Means discovered four meaningful customer behavioral groups without using any labels.

From Clusters to Action — Personalised Strategies

Our K-Means clustering has revealed distinct customer segments. Now, we translate these insights into concrete, actionable strategies tailored to each group, optimizing business outcomes.

Slow-Fulfilment Orders

Optimize logistics; offer express delivery options.

Budget-Conscious Buyers

Discount promotions, bundle deals

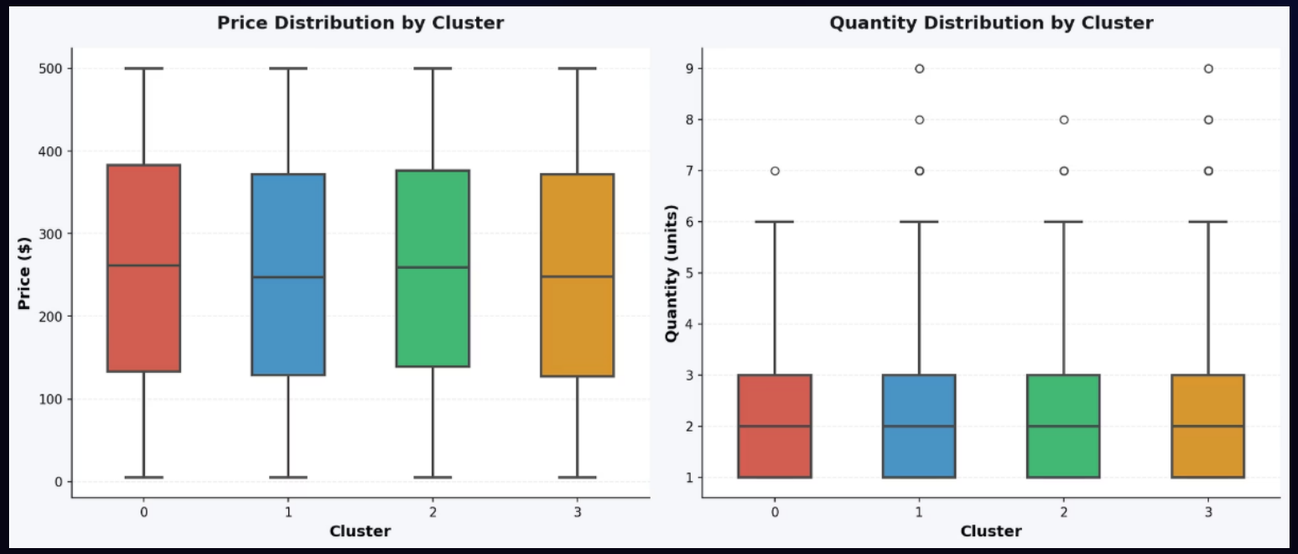
Premium / VIP Customers

Provide premium service, loyalty rewards, priority shipping.

Average / Returning

Implement retention campaigns and upsell opportunities.

These targeted actions foster cross-functional synergy, driving efficiency and customer satisfaction across the board.



Cross-Functional Impact

Marketing

Targeted campaigns per cluster

Operations

Shipping tier optimization

Finance

Dynamic pricing strategies

Customer Success

Personalised engagement

Customer segments inform each function, aligning teams around tailored actions that improve overall business outcomes.

Key Message: Each cluster maps to a concrete business action — this is actionable intelligence, not just data science.

AuraCart Sentinel — Intelligence On Duty

Final Performance Scorecard

Task	Model	Key Metric	Result
Price Prediction	Linear Regression	MAE	$98.94
Customer Segments	Gradient Boosting	Accuracy	81.80%
Delivery Status	Logistic Regression	Accuracy	70.45%
Customer Clustering	K-Means (k=4)	Clusters	4 found

Production Pipeline Summary

AuraCart Sentinel demonstrates a robust, end-to-end ML workflow:

- Tracked in MLflow (20+ experiments)
- Packaged as `.joblib` pipeline
- Deployed to Google Cloud Vertex AI
- Live predictions via REST API

> **Key Message:** AuraCart Sentinel delivers a complete, production-grade ML system from data exploration to cloud deployment.



Thank you.

> “Your questions drive our innovation forward.”

— CoreGenix Team